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European Patent Application 16802532.8
USE OF PTERIDINONE DERIVATIVE SERVING AS EGFR
INHIBITOR
East China University of Science and Technology

Responsive to the communication dated 28 June 2021:

An amended set of 8 claims is submitted in mark-up and clean copy which should be the basis for the further prosecution.

Claims 7 and 8 have been reworded in the "second medical use" format, based on PCT claims 7 to 9. The claims 7 and 8 were already amended to this format with the response dated 21 June 2019, but apparently, the amendment was omitted in the previous responses.

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Inventive step

The Examining Division has already acknowledged novelty of the presently claimed subject matter as well as an inventive step for subject matter wherein "R¹ is substituted or unsubstituted phenyl, optionally substituted C₃-C₈ cycloalkyl".

However, it was objected to that the three extra compounds, which have a hydrogen at the position of R¹, and which were included in claim 1 with the previous response, are not based on an inventive step.

It is argued that the three extra compounds 4, 5 and 12 are worse in their inhibitory activity than the structurally closest compound 32 of Table 2 of D4. According to the Examining Division, the compound 32 of D4 inhibits the proliferation of H1975 cells with an IC₅₀ value of 0.0039 µM. However, it appears that this argument is based on a misunderstanding of Table 2 of the Chinese publication D4. Table 2 on pages 30 and 31 of the Chinese document D4 shows the IC₅₀ values in µM on proliferation of H1975 cells. As becomes clear from page 31, line 5 of D4, the IC₅₀ value of compound 32 on H1975 cells is **0.038 µM**, and not 0.0039 µM as stated in the office action.

Table 1 of the present application shows the inhibitory activity on proliferation of H1975 cells of the compounds 4, 5 and 12 (which are the extra compounds of claim 1). The IC₅₀ values are 0.009, 0.006 and 0.006 µM, respectively. Therefore, the compounds 4, 5 and 12 are significantly better in their inhibitory activity than compound 32 of D4.

The compounds 4, 5 and 12 show excellent EGFR inhibitory activity. The compounds show a significantly higher inhibitory activity on the proliferation of H1975 cells than the structurally related compound 32 of D4. Starting from D4, it was completely unpredictable that compounds 4, 5 and 12 would show such excellent inhibitory activity on proliferation of H1975 cells.

Nothing in D4 or in the other prior art documents hinted at the significant increase of inhibitory activity on proliferation of H1975 cells which could be obtained by the structural modifications of compounds 4, 5 and 12. Therefore, the three extra compounds in claim 1 are based on an inventive step.

The subject matter of the present set of claims is based on an inventive step.

Adaption of the description

Replacement pages of the complete description are submitted in mark-up and clean copy.

Prior art documents D1 to D4 have been identified and briefly discussed on replacement pages 2 and 3.

It has been clarified on replacement page 3 that the invention is defined in the claims and that aspects not falling within the scope of the claims are disclosed and for information only.

Embodiments not falling within the scope of the claims have been labelled as "not covered by the claimed invention" or as "comparative".

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The statements "all references mentioned in the present application are incorporated herein by reference..." and "...such equivalent forms also fall within the scope of the appended claims..." have been deleted.

The description was broad into conformity with the present set of claims.

It is kindly requested to issue the communication pursuant to Rule 71(3) EPC.

Dr. Christoph Schreiber
European Patent Attorney

Enclosures: amended claims (mark-up)
 amended claims (clean copy)
 amended description (mark-up)
 amended description (clean copy)